

# NEXORA AUTHENTIC PREVIOUS YEAR QUESTIONS

**Exam:** UPSC Combined Geo-Scientist  
**Subject:** Geology  
**Questions:** 248  
**Source:** Official source PDF extraction  
**AI-generated PYQs:** 0

## 1. [UNKNOWN] paper-i

Q2. Explain the geological and palaeontological evidences of continental drift and comment on the mechanism of plate separation and collision. 15 Using case studies, explain the application of geomorphology in mineral prospecting. 15 Describe unconformities and their types, and give the various criteria to distinguish them from faults. 10

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## 2. [UNKNOWN] paper-i

Q3. Discuss the evidences of seafloor spreading and explain with illustrative diagrams, how earthquakes and subduction processes are related to each other. 15 Describe the components and working principles of (i) Cross-track scanners, and RISAT. 15 Enumerate the tools and equipment required to undertake geological mapping. Explain the importance of geological mapping. 10

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## 3. [UNKNOWN] paper-i

Q4. With the help of a case study, describe the methodology for using satellite imagery and GIS to identify potential zones of groundwater. 15 How are stress and strain related to each other? Explain the stress — strain curves in terms of brittle, ductile and viscous materials. 15 Discuss the internal structure of the Earth and comment on S and P-waves shadow zones with suitable diagrams. 10 FSI-P-GLY 2 ===== PAGE 3 ===== SECTION B

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## 4. [UNKNOWN] paper-i

Q5. Write explanatory notes within 150 words each, on the following: 8x5=40 Mineral-walled microfossils and their applications in stratigraphic correlation 8 Chronological order of volcanic events of the Indian Plate, and explanation of causes and manner of Deccan volcanism 8 Ghyben-Herzberg relationship 8 Palaeogeographic reconstruction of India during Gondwana times, using suitable diagrams 8 Causes, effects and mitigation measures of concrete cancer 8

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## 5. [UNKNOWN] paper-i

Q6. (a) Define Lithostratigraphy and comment on the criteria for lithostratigraphic classification and description of various rank terms. 15 Briefly describe organic-walled microfossils. Discuss their applications in petroleum exploration. 15 Describe the geological considerations for designing and constructing earthquake resistant structures. /0

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## 6. [UNKNOWN] paper-i

Q7. (a) Elaborate the procedure for 'Aquifer Performance Test' and estimation of hydraulic properties of aquifers. Describe using diagrams, the morphology of Graptolites and comment on the significance of Graptolites in the evolution of vertebrates. Discuss the stratigraphy and distribution of marine depositional episodes in the Gondwana sequence of India.

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## 7. [UNKNOWN] paper-i

Q8. (a) Discuss the geological history of Trilobites with specific description of the abundance, decline and extinction of the group through the Palaeozoic Era. Give a detailed account of the field and laboratory tests carried out to determine the engineering properties of rocks. Tabulate the Palaeozoic stratigraphic successions and their equivalents in Spiti, Kashmir and Kumaon basins of the Himalayas. FSI-P-GLY 3 15 15 10 15 15 10

### 8. [UNKNOWN] paper-ii

1. The principle of uniformitarianism states that: (a) The same external and internal processes that we recognize today have been operating consistent throughout most of earth's history (b) The rate of external and internal processes that we recognize today have been operating consistent throughout most of earth's history (c) The rate of rock cycles have been constant throughout the earth's history (d) The rate of rock cycles have changed through the different periods of earth's history

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### 9. [UNKNOWN] paper-ii

2. Which one of the following is the most widely accepted model of the "evolutionary theories of solar system"? (a) Nebular hypothesis (b) Planetesimal hypothesis (c) Gaseous tidal hypothesis (d) Gas dust cloud hypothesis

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### 10. [UNKNOWN] paper-ii

3. The boundary between the sial and sima is called: (a) Gutenberg discontinuity (b) Mohorovicic discontinuity (c) Conrad discontinuity (d) Gutenberg-Weichert discontinuity

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### 11. [UNKNOWN] paper-ii

4. The main cause of blowing wind across the earth surface is: (a) Wind blows from oceans to the land masses under the influence of sea-motion (b) Heating of atmosphere, land and oceans, accompanied by changes in air pressure (c) Wind blows from the higher elevation to the lower elevation derived by elevation difference (d) Wind blows from low pressure area to high pressure area due to pressure difference ===== PAGE 2 =====

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### 12. [UNKNOWN] paper-ii

5. Write short notes on the followings: 5x8=40 Red Sea Mineralisation. 8 Supergene alteration of sulphide deposits. 8 Placer deposits. 8 Coordination number and its relation to the cation-anion radius ratio in a crystal structure. 8 Uttarakhand natural disaster of June 2013. 8 What are the characteristics of Volcanics-associated Massive Sulphide (VMS) deposits? Discuss about the mineralogy and texture of VMS deposits. 10 How does coal form in nature? Why economic deposit of coal is not expected to develop in Precambrian formations? Briefly describe the distribution of Gondwana coal in Peninsular India. 10 Trace elements can be classified into compatible and incompatible elements. What is the basis of such classification? 10 Describe the important aspects of geochemical prospecting. 10 3 fsi-d-gly ===== PAGE 4 ===== • What are the major minerals of manganese? Write a note on manganese deposits of Sausar Group, India. 10 Why Banded Iron Formations (BIF) are restricted to Precambrian time? Enumerate the characters of Lake Superior type BIF. 10 How does volcanic activity affect climate? What types of volcanoes are more hazardous and how? 10 Describe the basis for open cost and underground mining. 10 What are the major ore minerals of aluminium and why do their composition vary with time? Discuss about the controls of ore localisation of bauxite through residual concentration. 10 What are the characteristic features to recognize any Proterozoic-type carbonate-hosted Pb-Zn deposit? Write a note on the sulphide deposits of Rajpura-Dhariba, Rajasthan. 10 How does the urbanization affect groundwater recharge? How can the recharge be increased? 10 What is a radioactive waste and why is it hazardous? Discuss the waste isolation versus waste disposal strategies for the radioactive waste. 10 fsi-d-gly 4

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### 13. [UNKNOWN] paper-ii

5. Megathrust earthquakes are the most damaging and deadly of all earthquakes, these megathrusts are located: (a) At the continent rift and transform plate boundary (b) In the divergent plate boundary zone (c) Along the contact between the subducting plate and the overriding plate (d) Along the contact between mid-oceanic ridges and transform fault

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#### 14. [UNKNOWN] paper-ii

6. The term 'Somma' refers to: (a) The highly saturated chlorous volcanic compounds comprising sodium, potassium and iron (b) The rim of the semi-destroyed old volcanic crater, inside which another cone may rise by subsequent eruptions (c) The enlargement of crater of volcanoes by various eroding agents (d) The depression on the slopes of volcanic cones

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#### 15. [UNKNOWN] paper-ii

7. Mid-ocean ridges contain 'black smokers' which are: (a) Volcanic smoke that contains mainly carbon in particulate form under deep sea (b) Hydrothermal vents from where hot water laden with fine grained sulfide minerals jets out into the cold sea water (c) Deep seated vents that pours out magma continuously in small amount (d) Deepest part of mid oceanic ridges which are largely inactive

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#### 16. [UNKNOWN] paper-ii

8. Trench rollback is: (a) Subduction of continental plate into asthenosphere and bounce back by the magmatic pressure (b) Spreading of oceanic plate and subduction into another trench (c) Subduction of oceanic plate into asthenosphere and bend toward the island arc (d) Bending of the subducting oceanic plate away from the island arc at deeper levels

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#### 17. [UNKNOWN] paper-ii

9. In glacier, the debris which is carried at the surface of ice is termed as: (a) Abrasion (b) Englacial transport (c) Supraglacial transport (d) Subglacial transport ===== PAGE 3 =====

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#### 18. [UNKNOWN] paper-ii

10. During the process of sand transport by aeolian action, the mechanical wear and tear of the sand particles among themselves is known as: (a) Abrasion (b) Attrition (c) Sandblasting (d) Deflation

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#### 19. [UNKNOWN] paper-ii

11. A stream emerging from a steep, narrow valley onto a broader lowland may build up a gently sloping conical deposit with an apex at the mouth of the narrow valley. Such a deposit is called: (a) Topset beds (b) Delta (c) Foreset beds (d) Alluvial fan

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#### 20. [UNKNOWN] paper-ii

12. Mineral quartz survives as sand grains in beaches, sand dunes, river and soil because: (a) It is the first mineral to crystallize in Bowen's reaction series, so it is an abundant mineral in the earth crust (b) It is resistant to chemical weathering as it crystallizes late according to Bowen's reaction series (c) It is resistant to chemical weathering as it allows easy substitution in its crystal structure (d) It is resistant to physical weathering, as it survives under frost wedging in low temperature condition

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#### 21. [UNKNOWN] paper-ii

13. The angular shear in a deformed rock body is described as: (a) Total degree of deflection of two lines which were originally at an certain angle before deformation (b) Angular relationship between alignments of two points change after a distortion deformation (c) The degree to which two originally perpendicular lines are deflected from 90° (d) The angle change between two parallel lines after the deformation ===== PAGE 4 =====

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## 22. [UNKNOWN] paper-ii

14. The appropriate mathematical expression for total strain (S) in a creep is: (where S = total strain, t = time, A, B and C are constants, D = strain during tertiary creep) (a)  $S = A + B e^{-Ct} - D$  (b)  $S = A + B e^{-Ct} + D$  (c)  $S = A - B e^{-Ct} + D$  (d)  $S = A + B e^{-Ct} + D$

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## 23. [UNKNOWN] paper-ii

15. Class-1 folds are distinguished by: (a) Dip isogons that converge downward, signifying that the curvature of the outer arc is less than that of inner arc (b) Parallel folds with curvatures of the outer arc and inner arc of the fold being identical (c) Fold marked by dip isogons that diverge downward as outer arc curvature exceeds inner arc curvature (d) Folds marked by modest thickening on the hinge and modest thinning on the limbs

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## 24. [UNKNOWN] paper-ii

16. 'Throw' in a fault is defined as: (a) Angle between the fault plane and a horizontal plane (b) Angle between strike of a fault plane and strike of regional structure (c) Angle between fault plane and a vertical plane (d) Thickness of stratigraphic interval between two beds that are brought into contact by faulting

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## 25. [UNKNOWN] paper-ii

17. Which of the following type of folds are classified on the basis of mechanism of folding? (a) Flexure-slip folds (b) Plunging and reclined folds (c) Symmetrical and recumbent folds (d) Parallel and similar folds

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## 26. [UNKNOWN] paper-ii

18. The assembly of tilted blocks, half grabens, grabens and horsts is called: (a) Strike-slip landscape (b) Pull-apart basins landscape (c) Fault-block landscape (d) Gully landscape ===== PAGE 5 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

## 27. [UNKNOWN] paper-ii

19. The value of normal stress and shear stress on each fault plane ( $\sigma_n$ ) and shear stress ( $\tau$ ) on each fault plane in the Mohr envelope of failure is reflected by: (a) The coordinates of the point making the intersection of each circle with its radius (b) The tangential reflection of each circle with X-axis (c) The coordinates of the point making the intersection with its applied stress (d) The tangential projection of  $\sigma_1$  with applied stress along Y-axis

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## 28. [UNKNOWN] paper-ii

20. Riecke's principle of recrystallization for plastic deformation is expressed as: (a) The rock crystallized according to orientation of their glide plane (b) The intergranular recrystallization takes place because of intergranular displacement between the individual grains (c) The preferred orientation in rocks are the product of guided recrystallization under high confining pressure and temperature (d) The solution in the pore spaces of the rocks dissolve that portion of the crystal which is under greatest stress and at the same time there is precipitation on that portion of the crystal which is subjected to the least stress

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## 29. [UNKNOWN] paper-ii

21. Cataclastic flow is the result of: (a) Dislocation by the pressure solution (b) Displacement along the dislocation planes (c) Flow along the crystallographic planes of weakness (d) Frictional sliding along the interconnected microcracks

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**30. [UNKNOWN] paper-ii**

22. The outcrop pattern of beds that dip downstream at an angle less than the gradient of the stream forms a 'V' which points: (a) In upstream direction (b) In downstream direction (c) Perpendicular to the direction of river flow (d) In strike direction of the beds parallel to flow of river ===== PAGE 6 =====

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**31. [UNKNOWN] paper-ii**

23. In a structural contour map 'homoclines' are distinguished by: (a) Inclined contour lines that cross cut a bedding plane with regular elevation interval (b) Subparallel contour lines that steadily decrease or increase in elevation value across the face of the map (c) Subparallel contour lines that always runs orthogonal to the bedding planes (d) Parallel contour lines that also run parallel to the bedding planes

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**32. [UNKNOWN] paper-ii**

24. Rake can be defined as: (a) Geometrical relationship between the plane in which line is found with respect to regional structures (b) Angle between the plane and horizontal datum line (c) Angle between a line and the strike line of the plane which contains that line (d) Angle between a line and a plain

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**33. [UNKNOWN] paper-ii**

25. Which one of the following coal constituents is hard, compact and has granular texture? (a) Vitrain (b) Durain (c) Clarain (d) Fusain

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**34. [UNKNOWN] paper-ii**

26. If the indices of any crystal face are rational numbers and are determined by dividing the intercepts of any face into the intercepts of the parametral plane and clearing fraction, that with intercepts a, b and c. The indices will be: (a) (101) (b) (111) (c) (001) (d) (100)

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**35. [UNKNOWN] paper-ii**

27. Bauxite is a mixture of: (a) Aegirine and Jadeite (b) Steatite, Aphrite, Mullite and Enstatite (c) Gibbsite, Diaspore and Boehmite (d) Tremolite, Glaucophanite and Riebeckite ===== PAGE 7 =====

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**36. [UNKNOWN] paper-ii**

28. A few non-metallic minerals are often found associated with an ore mineral. These have to be separated from the ore before the same is processed for extraction of the metal. These associated minerals are called: (a) Ore minerals (b) Gangue minerals (c) Syngenetic minerals (d) Supergene minerals

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**37. [UNKNOWN] paper-ii**

29. Which one of the following group of minerals are classified within sulphides? (a) Bromides (b) Fluorides (c) Arsenates (d) Arsenides

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**38. [UNKNOWN] paper-ii**

30. Opal is a mineral that displays iridescence or play of colours when rotated or observed from different direction because of: (a) Interference of the ray of light by minute globules of water trapped in the outer layer of crystal lattice (b) Interference of the ray of light by the distortions in crystal lattice (c) On account of mineral inclusions (d) Interference of light reflected from thin plates of other minerals enclosed or exsolved in parallel planes within the crystal

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### 39. [UNKNOWN] paper-ii

31. "Pleochroism" representing change of colours upon rotation of microscope stage under plane polarized light in some minerals is characterized by: (a) Two extremes of colour seen twice during a complete 360° rotation (b) Four extremes of colour seen twice during a complete 360° rotation (c) Two extremes of colour seen twice during a 90° rotation (d) Four extremes of colour seen twice during a 90° rotation ===== PAGE 8 =====

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### 40. [UNKNOWN] paper-ii

32. In alkali feldspar a 'coupled substitution' occurs when: (a) Addition of one silicon ion in the framework structure to make a covalent bond in alkali feldspar (b) One aluminum ion plus one sodium ion enter the framework structure and replace one silicon ion and, in addition, fill a vacant site (c) One silicon ion is replaced by the one sodium ion to balance one ion of oxygen (d) One oxygen ion in addition enter in the framework structure to balance the charge of silicon and alumina

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### 41. [UNKNOWN] paper-ii

33. Garnet in-situ deposits are generally associated with: (a) Sedimentary rocks only (b) Igneous rocks only (c) Metamorphic rocks (d) Sedimentary and igneous rocks

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### 42. [UNKNOWN] paper-ii

34. Variation in the refractive index in  $\text{Al}_2\text{SiO}_5$  polymorphs is due to: (a) Polarization of some light in the form of interference colour (b) Ferric iron and manganese entering structure (c) Thickness of thin section that retard velocity of light (d) Transmission of light with equal velocity in all direction

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### 43. [UNKNOWN] paper-ii

35. The ray velocity surface of an anisotropic crystal is an ellipsoid because anisotropic crystals: (a) Transmit light with same velocities in different directions (b) Transmit light with different velocities in different directions (c) Absorb all the transmitted light (d) Have lower refractive index than air ===== PAGE 9 =====

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### 44. [UNKNOWN] paper-ii

36. Mineral Quartz shows twinning according to Brazil law (twin plane  $112\bar{0}$ ) and Dauphiné law (twin axis is c-axis) but this twinning is not detectable under the microscope because: (a) Twin plane ( $112\bar{0}$ ) is always perpendicular to the twin axis (c-axis) (b) Both  $112\bar{0}$  and c-axis are in extinction position in twin plane direction (c) Optic orientation in both twin parts is identical in both types of twin (d) Optic orientation in both twin parts is not identical in both types of twin

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### 45. [UNKNOWN] paper-ii

37. When basaltic lava enters into water bodies, it tends to form large sack-like bodies called: (a) Pahoehoe lavas (b) 'aa' lavas (c) Pillow lavas (d) Lahars

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### 46. [UNKNOWN] paper-ii

38. Igneous rocks formed by gravity settling of the crystals in the magma chamber floor represent: (a) Pilotaxitic texture (b) Variolitic texture (c) Cumulophyric texture (d) Cumulate texture

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### 47. [UNKNOWN] paper-ii

39. Which one of the following rocks contain nearly equal amount of alkali feldspar and plagioclase with approximately

10% quartz? (a) Quartz syenite (b) Quartz monzonite (c) Quartz monzodiorite (d) Monzosyenite

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**48. [UNKNOWN] paper-ii**

40. During fractionation, separated crystals of pyroxene (augite) and intermediate plagioclase feldspar form a crystallized rock known as: (a) Websterite (b) Lherzolite (c) Pyroxenite (d) Gabbro ===== PAGE 10 =====

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**49. [UNKNOWN] paper-ii**

41. Rhyolitic magmas tend to rise in large dome-like bodies known as: (a) Plinian plumes (b) Diatremes (c) Diapirs (d) Phreatoplinian

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**50. [UNKNOWN] paper-ii**

42. When cooling of mineralizing liquid is slow, leucite inverts to kalsilite-K-feldspar intergrowth. The intergrowth is termed as: (a) Perthite (b) Polyleucite (c) Pseudoleucite (d) Antiperthite

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**51. [UNKNOWN] paper-ii**

43. In layered intrusions, variation in mineral composition is normally not evident until analysis has been done; such layering is called: (a) Modal layering (b) Comb layering (c) Cryptic layering (d) Ristic layering

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**52. [UNKNOWN] paper-ii**

44. Desilication reaction occurs when a: (a) Carbonate sedimentary rock reacts with silicate magma and produces calcium magnesium silicates (b) Primary alkaline magma generated from primitive sub-alkaline basaltic magma (c) Low pressure crystal fractionation from sub-alkaline basaltic magma (d) Silica under-saturated primary magma generated at relatively high pressure and CO<sub>2</sub> concentration

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**53. [UNKNOWN] paper-ii**

45. The magnesium number designated by 'M' is particularly used as: (a) Olivine control line (b) Evaluation of depleted or enriched mantle (c) Partial melting or enriched event (d) Differentiation index ===== PAGE 11 =====

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**54. [UNKNOWN] paper-ii**

46. Nephelinite magma crystallizes to form which one of the following suite of igneous rocks? (a) Vogesite (b) Ijolite (c) Limbergite (d) Mugearite

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**55. [UNKNOWN] paper-ii**

47. Which one of the following correctly represents the composition of kimberlite? (a) Nickel and calcium rich rock with garnet and biotite phenocrysts (b) Volatile-rich potassic ultramafic rock composed of phenocrysts of olivine and phlogopite (c) Sodium rich rock with high amount of biotite and hornblende (d) Dark coloured iron and magnesium rich rock associated with mid-oceanic ridges

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**56. [UNKNOWN] paper-ii**

48. During rapid quenching when phenocryst is surrounded by glass, the texture is termed as: (a) Aphyric (b) Vitrophyric (c) Cumulophyric (d) Glomerophyritic

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**57. [UNKNOWN] paper-ii**

49. Which one of the following is similar to slate but has a silky sheen on the cleavage surface? (a) Helicitic (b) Phyllite (c) Quartzite (d) Gneiss

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**58. [UNKNOWN] paper-ii**

50. If the foliation is caused by a parallel arrangement of medium to coarse grained phyllosilicate minerals, it is called: (a) Poikiloblastic (b) Lineation (c) Helicitic (d) Schistosity ===== PAGE 12 =====

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf>

**59. [UNKNOWN] paper-ii**

51. With increasing metamorphic intensity, a granoblastic texture results in: (a) Increase in grain size (b) Decrease in grain size (c) No variation in grain size (d) Loss of fabric

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf>

**60. [UNKNOWN] paper-ii**

52. Armoured relics are: (a) Minerals as inclusion even though they are no longer present in the rock matrix (b) High density of inclusion similar to matrix mineral composition (c) Porphyroblasts with no inclusions (d) Replacement of pre-existing mineral grains by inclusion minerals

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf>

**61. [UNKNOWN] paper-ii**

53. In a 'snowball crystal' if internal schistosity is undisturbed in the centre of porphyroblast but becomes increasingly tightly folded towards the margins during the formation of crenulation cleavage, this feature represents: (a) Syntectonic growth pattern (b) Post-tectonic growth pattern (c) Pre-tectonic growth pattern (d) Increasing volume growth pattern

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**62. [UNKNOWN] paper-ii**

54. In metamorphic rocks, when all grains are of the same size and have planar boundaries which are intersecting at approximately 120°, then the texture is known as: (a) Granoblastic polygonal texture (b) Decussate texture (c) Undulose extension (d) Ribbon texture ===== PAGE 13 =====

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf>

**63. [UNKNOWN] paper-ii**

55. In a zoned olivine crystal, Fe tends to diffuse away from the fayalite-rich portions towards the parts enriched in forsterite, while Mg will tend to diffuse in the opposite direction. This type of diffusion is known as: (a) Grain boundary diffusion (b) Flux diffusion (c) Self diffusion (d) Interdiffusion

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf>

**64. [UNKNOWN] paper-ii**

56. Which one of the following is defined as a line joining points where the rock have suffered metamorphism under similar pressure-temperature conditions? (a) Isograd (b) Grade (c) Facies (d) Zones

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf>

**65. [UNKNOWN] paper-ii**

57. An 'eclogite' is: (a) A green coloured rock dominated by pyroxene with plagioclase (b) Red and green, dense rock, dominated by garnet and omphacitic pyroxene and lacking plagioclase (c) Red and green dense rock dominated by plagioclase, garnet and pyroxene (d) A rock with garnet and pyroxene with abundance of mica

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf>

**66. [UNKNOWN] paper-ii**

58. The granulite facies rocks that at high grades develop felsic segregations due to partial melting are called: (a)



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**67. [UNKNOWN] paper-ii**

59. The characteristic of granulite facies is: (a) Formation of hydrous Ca-Al silicate with zeolite, prehnite and pumpellyite (b) Presence of mineral garnet with metabasites (c) Extreme presence of Glaucophane without quartz (d) Development of both clinopyroxene and orthopyroxene commonly with quartz and plagioclase

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**68. [UNKNOWN] paper-ii**

60. Which one of the following represents the location of metamorphic paired belts? (a) Along the divergent plates (b) Subduction zone beneath island arc and continental margins (c) Along transform plates (d) Along mobile belts

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**69. [UNKNOWN] paper-ii**

61. Sediments which contain grains of various sizes in nearly equal amount are said to be: (a) Well assorted (b) Unassorted (c) Positively skewed (d) Composite

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**70. [UNKNOWN] paper-ii**

62. Which one of the following is a variety of sandstone that is exceptionally rich in mica dispersed in parallel or subparallel layers? (a) Arkose (b) Flagstone (c) Freestone (d) Ganister

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**71. [UNKNOWN] paper-ii**

63. A muddy, laminated and fissile rock capable of breaking into small flakes is called: (a) Augen gneiss (b) Sand stone (c) Micaceous quartzite (d) Shale ===== PAGE 15 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**72. [UNKNOWN] paper-ii**

64. Which one of the following explains the chelation process of weathering? (a) Breakdown of mineral due to the extraction of water molecules during the process of weathering (b) Bonding of metal ions to organic substances to form organic molecules and removal of cations from mineral lattice and keeping the cations in solution (c) Addition of water molecules into the mineral to form a new mineral (d) Chemical breakdown of mineral by the process of thermal expansion

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**73. [UNKNOWN] paper-ii**

65. On account of the charged surfaces of clay minerals, their interaction with saline water leads to clumping into aggregates of randomly oriented clay flakes. This process is termed as: (a) Fissility (b) Flocculation (c) Lamination (d) Illitization

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**74. [UNKNOWN] paper-ii**

66. Which one of the following beddings is markedly trough or scoop-shaped? (a) Festoon bedding (b) Massive bedding (c) Lenticular bedding (d) Laminated bedding

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**75. [UNKNOWN] paper-ii**

67. Which one of the following is a grain-supported limestone that also contains some micrite mud matrix? (a) Grainstone (b) Packstone (c) Wackestone (d) Mudstone

**76. [UNKNOWN] paper-ii**

68. During diagenesis, the process of intrastratal solution leads to: (a) Dissolution of one mineral and precipitation of another mineral simultaneously (b) Selective dissolution of less stable framework grains or part of grains (c) Alteration of one mineral into other by the process of dewatering (d) Formation of two minerals of contrasting physical and chemical environment ===== PAGE 16 =====

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**77. [UNKNOWN] paper-ii**

69. Which one of the following rock types may also be formed along planes of movement in the crust? (a) Conglomerates (b) Limestone (c) Breccia (d) Gravel

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**78. [UNKNOWN] paper-ii**

70. Rip currents that occur near beach are localized in nature and: (a) Move seaward as a narrow, near surface current (b) Move in turbulent manner in shoaling zone with reduced velocity (c) Are the perennial currents in the sea driven by the circulation pattern of earth (d) Are deflection currents which play an important role in dispersal of sediment in wide area along beaches

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**79. [UNKNOWN] paper-ii**

71. The replacement of dolomite rhombs by calcite is termed as: (a) Dolomitization (b) Dedolomitization (c) Recrystallization (d) Substitution

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**80. [UNKNOWN] paper-ii**

72. The Dead Sea provides: (a) A single modern example of a deeper water evaporite environment (b) Most remarkable deposit of evaporite from the ancient past (c) Most modern evaporite deposit of marine sabkhas (d) An example of ancient sea which has disappeared now

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**81. [UNKNOWN] paper-ii**

73. The organism is decomposed and it loses its nitrogen, oxygen and other volatile constituents, this process is known as: (a) Petrification (b) Carbonisation (c) Moulds (d) Impression ===== PAGE 17 =====

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**82. [UNKNOWN] paper-ii**

74. The study of the process of fossilization is called: (a) Agronomy (b) Teronomy (c) Taxonomy (d) Taphonomy

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**83. [UNKNOWN] paper-ii**

75. Paratype is a specimen, other than the holotype, which is formally designated by the author of a species as: (a) Having been used in the description of the species (b) A syntype which is subsequently singled out as the definitive type specimen for a species (c) Used for re-description of existing species (d) Depicting it as non-original type material which has been collected at the type locality

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**84. [UNKNOWN] paper-ii**

76. In paleontology, subspecies is usually used to denote: (a) The extinction of one species by the expense of one species forms a subspecies (b) The morphological changes in a species that lead to transformation of one species to another (c) The evolution of new species from the mutation of old genera (d) The intermediate stage in the replacement

of one species by another through phyletic transition

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**85. [UNKNOWN] paper-ii**

77. Which one of the following statements regarding bivalve is correct? (a) Length is defined as the maximum linear dimension in an antero-posterior direction and height is defined as the maximum dimension perpendicular to the length (b) Length is defined as the maximum linear dimension in the radial direction while height is defined as the maximum dimension perpendicular to the length (c) Length is defined as the maximum linear dimension in posterior-anterior direction while height is perpendicular to the length (d) Length is the maximum area covered by bivalve shell in one linear direction while height is perpendicular to it ===== PAGE 18 =====

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**86. [UNKNOWN] paper-ii**

78. The body part of which invertebrate can transversely and longitudinally be divided into distinct three parts? (a) Brachiopods (b) Gastropods (c) Trilobites (d) Cephalopods

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**87. [UNKNOWN] paper-ii**

79. Brain size in different mammals generally tends to increase during the process of evolution by the powers of: (a)  $2/3$  of body size (b)  $3/2$  of body size (c)  $4/3$  of body size (d)  $1/6$  of body size

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**88. [UNKNOWN] paper-ii**

80. Which one of the following is related to the 'missing link' between humans and their ape like ancestors? (a) Proconsul africanus (b) Homo habilis (c) Homo erectus (d) Homo heidelbergensis

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**89. [UNKNOWN] paper-ii**

81. The genus Orohippus which represents the second stage in evolution of horse is popularly known as: (a) Dawn horse (b) Mountain horse (c) Browsing horse (d) Siwalik horse ===== PAGE 19 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**90. [UNKNOWN] paper-ii**

82. Large benthic foraminifera became abundant in diversity and quantity since: (a) Cretaceous (b) Eocene (c) Oligocene (d) Carboniferous

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**91. [UNKNOWN] paper-ii**

83. Which one of the following plant fossils exhibit lanceolate pinnules type leaf with prominent mid-rib and veins? (a) Ptilophyllum (b) Glossopteris (c) Lepidodendron (d) Thinnfeldia

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**92. [UNKNOWN] paper-ii**

84. Which one of the following foraminifera fossils is found in Ladakh and Kashmir? (a) Fusulina (b) Globotruncana (c) Globorotalia (d) Lituotuba

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**93. [UNKNOWN] paper-ii**

85. The Permian rock formation of the Salt Range in Pakistan (Productus Limestone) is lithologically different from that of the Spiti area (Productus Shale) while having same relative age because: (a) Both the rock formations were deposited in same sedimentary basin but were altered later by different grades of deformation (b) Both the rock formations were deposited in different sedimentary basins in spite of the same relative age (c) In Pakistan, the

Productus Limestone was deposited as vertical succession during sea transgression while in the Spiti area Productus Shale was deposited as lateral succession (d) In Spiti area, the Productus Shales have been deposited over the Productus Limestone and both the formations belong to different age groups ===== PAGE 20 =====

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf>

**94. [UNKNOWN] paper-ii**

86. In stratigraphic correlation, the term homotaxial refers to the formations: (a) Deposited at the same stage of the earth's evolutionary history (b) Deposited at different geological time (c) Having similar lithologic package (d) Bearing same fossil content despite in different time

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf>

**95. [UNKNOWN] paper-ii**

87. The Singhbhum Shear Zone is also known as: (a) Singhbhum Zinc Belt (b) Singhbhum Copper Belt (c) Singhbhum Silver Belt (d) Singhbhum Gold Belt

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf>

**96. [UNKNOWN] paper-ii**

88. The stratigraphic boundary demarcating the Archaean rocks from Closepet granite is termed as: (a) Non-conformity (b) Disconformity (c) Angular Unconformity (d) Eparchaeon Unconformity

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf>

**97. [UNKNOWN] paper-ii**

89. The sediments of the 'Dharwar Super-group' were deposited over a basement of: (a) Slates and phyllites (b) Marbles and Conglomerates (c) Sargur Schists and Peninsular Gneissic complex (d) Basalts and conglomerates

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf>

**98. [UNKNOWN] paper-ii**

90. The Chipli Ghat Group comprises: (a) Huge thickness of phyllites, slates and mica-schists (b) Rich manganese deposit (c) High grade metamorphic rocks of plutonic origin (d) Garnet-sillimanite schists termed as Khondalite rocks  
===== PAGE 21 =====

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf>

**99. [UNKNOWN] paper-ii**

91. Vempalle shales and limestones belong to: (a) Kistna group (b) Nallamalai group (c) Cheyair group (d) Papaghni group

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**100. [UNKNOWN] paper-ii**

92. Which one of the following is the characteristic feature of extrusive and intrusive phase of 'Malani Igneous Suite'? (a) It comprise mainly mafic magmatic activity which is in the form of numerous dykes (b) It represents one of the world's largest example of felsic magmatism (c) It is mainly composed of layered structures (d) Its rocks resemble Archaean basement by chemical composition and are considered to represent oldest known magmatic activity

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf>

**101. [UNKNOWN] paper-ii**

93. The oldest group of Cuddapah Super-group is: (a) Papaghni Group (b) Nallamalai Group (c) Chitravati Group (d) Kurnool Group

Official source: <backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf>

**102. [UNKNOWN] paper-ii**

94. The Cretaceous rocks exposed along the southern slopes of the Shillong Plateau have been grouped into: (a) Mahadek Formation (b) Longpar Formation (c) Dalmiapuram Formation (d) Lathi Formation

**103. [UNKNOWN] paper-ii**

95. The worldwide orogenic event which took place during Ordovician and Silurian period is known as: (a) Hercynian orogeny (b) Caledonian orogeny (c) Catalatic orogeny (d) Mesonian orogeny ===== PAGE 22 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**104. [UNKNOWN] paper-ii**

96. The linear Ophiolitic belt of Ladakh was formed as a consequence of: (a) Thick sediment deposition by transgression of Tethys sea (b) High rate of erosion of the rising Himalaya due to the northward drift of Indian plate (c) Emplacement of acidic volcanics in deep sea in association of marine sediment (d) Emplacement of basic volcanics in association with the deep marine sediment

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**105. [UNKNOWN] paper-ii**

97. Which one of the following processes is most commonly associated with the fumaroles? (a) Metasomatism (b) Hypothermal (c) Sublimation (d) Mesothermal

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**106. [UNKNOWN] paper-ii**

98. Which one of the following is commonly known as "black lead" but, lacks lead in its composition? (a) Sphalerite (b) Graphite (c) Pentlandite (d) Cassiterite

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**107. [UNKNOWN] paper-ii**

99. Which one of the following minerals is used as an important raw material for the manufacture of cement and fertilizer-ammonium sulphate? (a) Barytes (b) Magnesite (c) Gypsum (d) Calcite

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**108. [UNKNOWN] paper-ii**

100. Which one of the following ore minerals is associated with the nepheline syenite in Khammam district of Andhra Pradesh? (a) Andalusite (b) Corundum (c) Sillimanite (d) Sheelite ===== PAGE 23 =====

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**109. [UNKNOWN] paper-ii**

101. Which silicate of iron and magnesium is termed as chrysolite? (a) Mica (b) Olivine (c) Magnesite (d) Biotite

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**110. [UNKNOWN] paper-ii**

102. Which one of the following statements regarding hydrothermal solutions is correct? (a) These are weakly concentrated, highly dissociated, basic solutions (b) These are weakly concentrated, highly dissociated, acidic solutions (c) These are concentrated, highly dissociated, electrolytic solutions (d) These are concentrated, weakly dissociated, electrolytic solutions

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**111. [UNKNOWN] paper-ii**

103. Which one of the following is a ladder vein type of deposit? (a) Iron deposits of Goa (b) Cassiterite deposit of Bastar, Chhattisgarh (c) Lead-Zinc deposit of Zawar, Rajasthan (d) Magnesite deposit of Mysore, Karnataka

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**112. [UNKNOWN] paper-ii**

104. A host of mineral deposits occur at the contact zone of meso and epi-zonal igneous plutons because: (a) Maximum

fracture/shearing and low thermal gradient exist here (b) Prominent lineament run parallel to the stratigraphic trends (c) These are weak zones, mixing of fluids takes place and thermal gradient is steep (d) Very intense weathering leads to the formation of residual deposits

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**113. [UNKNOWN] paper-ii**

105. Which one of the following is correct for the bituminous coal? (a) Its calorific value ranges between 11,000 to 12,500 B.T.U. (b) It shows banded structure with alternate dull and bright bands (c) It shows sign of original vegetal matter (d) It contains very high volatile matter ===== PAGE 24 =====

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**114. [UNKNOWN] paper-ii**

106. The organic theory of oil genesis appeared for the first time in the form of: (a) Gubkin's hypothesis (b) Lomonosov's distillation hypothesis (c) Pryor's distillation hypothesis (d) Bosworth's theory

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**115. [UNKNOWN] paper-ii**

107. Which one of the following properties of mica make it an excellent electrical insulator? (a) Its splitting property combined with flexibility, elasticity, resilience, low heat conductivity and high dielectric strength (b) Its hardness, lustre, malleability and electric strength (c) To easily spun into yarn and its resistance to heat (d) Its inert and stable nature

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**116. [UNKNOWN] paper-ii**

108. Petroleum and natural gas deposits in Assam are associated with rocks of: (a) Jurassic age (b) Precambrian age (c) Eocene age (d) Oligocene to Lower-Miocene age

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**117. [UNKNOWN] paper-ii**

109. Coalescence process is one by which: (a) Nucleation of raindrops develops around the organic or dust particles (b) The small cloud droplets increase their size due to contact with other droplets through collision (c) Sea water brought into the environment during the evaporation (d) The tiny rain drops are held in the environment just before the precipitation ===== PAGE 25 =====

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**118. [UNKNOWN] paper-ii**

110. Which one of the following is the correct equation for the calculation of the volume of water absorbed in a rock? (where,  $P$  = porosity,  $W_s$  = saturated weight,  $W_d$  = dry weight,  $V$  = volume,  $D$  = density) (a)  $P = 100 \times \frac{W_s - W_d}{W_s} \times \frac{D}{V}$  (b)  $P = 100 \times \frac{W_s - W_d}{W_d} \times \frac{D}{V}$  (c)  $P = 100 \times \frac{W_s - W_d}{W_s} \times \frac{V}{D}$  (d)  $P = 100 \times \frac{W_s - W_d}{W_d} \times \frac{V}{D}$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**119. [UNKNOWN] paper-ii**

111. The barometric efficiency (BE) of a confined aquifer derived by Jacob is expressed as: (where,  $M$  = bulk modulus of compression of the solid skeleton of the aquifer,  $M_w$  = bulk modulus of compression of water,  $P$  = porosity of the aquifer) (a)  $BE = 1 + \frac{M}{M_w}$  (b)  $BE = 1 - \frac{M}{M_w}$  (c)  $BE = 1 + \frac{M_w}{M}$  (d)  $BE = 1 - \frac{M_w}{M}$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**120. [UNKNOWN] paper-ii**

112. Which one of the following is called the secondary porosity? (a) The fractures in the igneous and metamorphic rocks developed due to tectonic activity (b) The primary pore spaces present in both unconsolidated and semi-consolidated sediment (c) The permeability of the rock which is directly proportional to the pore spaces and size of pore (d) The pore spaces that develop at the time of formation of a rock or sediment

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**121. [UNKNOWN] paper-ii**

113. The coefficient of transmissivity (T) can be expressed as: (a)  $T = 1 K \times b$  (b)  $T = cd^2$  (c)  $T = \sqrt{K} \times b$  (d)  $T = K \times b$   
===== PAGE 26 =====

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**122. [UNKNOWN] paper-ii**

114. The quantity of water (Q) moving per unit of time, as stated by Darcy's law is expressed as: (where, P = coefficient of permeability, I = hydraulic gradient, A = cross-sectional area through which water moves) (a)  $Q = PIA$  (b)  $Q = 1 PIA$  (c)  $Q = \sqrt{PIA}$  (d)  $Q = 1 \sqrt{PIA}$

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**123. [UNKNOWN] paper-ii**

115. According to Darcy's law, the rate of flow of water through a porous media is: (a) Inversely proportional to the difference in hydraulic head between the two ends of the porous media (b) Proportional to the difference in hydraulic head between the two ends of the porous media (c) Proportional to the length of the flow path (d) Inversely proportional to the area of cross section of the aquifer

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**124. [UNKNOWN] paper-ii**

116. The total mechanical energy per unit weight (h) as expressed by Bernoulli equation is: (where, P = pressure,  $\gamma$  = specific weight of water per unit volume, V = velocity of flow of water,  $\rho$  = density of water, g = acceleration due to gravity, Z = elevation head measured from a datum or sea level) (a)  $h = \frac{P}{\gamma} + \frac{V^2}{2g} + Z$  (b)  $h = \frac{P}{\rho} + \frac{V^2}{2g} + Z$  (c)  $h = \frac{P}{\gamma} + \frac{V^2}{2g} + Z$  (d)  $h = \frac{P}{\rho} + \frac{V^2}{2g} + Z$  ===== PAGE 27 =====

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**125. [UNKNOWN] paper-ii**

117. Which one of the following is the correct expression for determining the Reynold's number (R)? (a)  $R = \frac{\rho V D}{\mu}$  (b)  $R = \frac{\rho V D}{\mu}$  (c)  $R = \frac{\rho V D}{\mu}$  (d)  $R = \frac{\rho V D}{\mu}$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**126. [UNKNOWN] paper-ii**

118. Sodium salts are highly soluble in water and are released in solution during: (a) Dissolution action of limestone (b) Decay of quartz by chemical replacement (c) Weathering of plagioclase feldspar (d) Weathering of olivine minerals

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**127. [UNKNOWN] paper-ii**

119. The contour trenches with bunds which are constructed for the artificial recharge of ground water are: (a) Parallel to topographic contour with plantation to check rapid runoff (b) Perpendicular to rivulets where permeable formation are present (c) Parallel to rivulets where permeable formation are present (d) Perpendicular to topographic contour with plantation to check rapid runoff

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**128. [UNKNOWN] paper-ii**

120. Induced recharge is an indirect method of artificial recharge involving: (a) The pumping from an aquifer which is hydraulically connected with surface water bodies (b) A method in which pumping lowers the ground water level and cone of recharge is formed (c) A method where stream bed is connected to aquifer by clayey formation (d) A method in which there is build up of ground water storage

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Paper-II-Geology-Hydrogeology.pdf

**129. [UNKNOWN] unknown**

1. The 'Correlation' determines the age relationship between rock units or geological events in separate area, but the

'Physical Continuity' is: (a) To trace physically the source of rock unit between the two different places (b) The correlation of two rock units according to geological time scale (c) The lateral correlation of the rocks which are exposed in a wide gap physically (d) The correlation of two rock units on the basis of their same physical and chemical character

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**130. [UNKNOWN] unknown**

2. Consider the following regarding a rock sample:  $t_{1/2}$  is half life of the radioactive substance  $N_0$  is number of atoms of parent radioactive substance present today  $N$  is number of atoms of the daughter product The age of a rock sample is determined by which one of the following equations? (a)  $3.323 \log \frac{N_0}{N} \frac{1}{t_{1/2}}$  (b)  $3.323 \log \frac{N_0}{N} \frac{1}{t_{1/2}}$  (c)  $3.323 \log \frac{N_0}{N} \frac{1}{t_{1/2}}$  (d)  $3.323 \log \frac{N_0}{N} \frac{1}{t_{1/2}}$  ===== PAGE 2 ===== 2

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**131. [UNKNOWN] unknown**

3. In "humid temperate climate" the rocks are weathered predominantly by a combination of both disintegration and decomposition, resulting in the formation of: (a) Black cotton soil (b) Lateritic soil (c) Podzol soil (d) Sandy soil

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**132. [UNKNOWN] unknown**

4. The molten spheroidal mass that the earth originally was started cooling with the result that a solid crust was formed over a molten core. The process of gradual release of previously dissolved gasses, in molten magma, is called: (a) Degassing (b) Diagenesis (c) Decomposition (d) Disintegration

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**133. [UNKNOWN] unknown**

5. The volcanism of island arcs and continental-margin orogens are associated with plate convergence. More extensive vast sheets of basaltic lavas which have formed on the continents at various times in the past are referred to as: (a) Continental flood basalts (b) Mid-oceanic lava flow (c) Andesitic lava flow (d) Rhyolitic lava flow ===== PAGE 3 ===== 3

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**134. [UNKNOWN] unknown**

6. Isostatic adjustment is: (a) The horizontal movement of crustal rock tending to rise or sink gradually until they are balanced by the weight of displaced mantle rocks. (b) The vertical movement of crustal rock tending to rise and sink gradually until they are balanced by the weight of displaced mantle rocks. (c) Shrinking of crust rock during the process of convergent plate boundary movement which is equal to applied stress (d) Rearrangement of morphological features at the surface of earth after removal of applied pressure

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**135. [UNKNOWN] unknown**

7. The 'truncated spurs' are the Erosional landscapes associated with valley glacier, these are: (a) Sharp peaks that remain after cirques have cut back into a mountain on several sides (b) Frost wedging on the rocks exposed above the glacier, steepening and cutting back the side wall of the valley (c) Lower parts of ridges that have been carved into triangular facets by glacial erosion (d) Steep sided, rounded hollow carved into a mountain at the head of a glacial valley

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**136. [UNKNOWN] unknown**

8. The quantity of water which enters stream channels in a drainage basin over a specified period of time can be determined by a: (a) Channel steepness equation (b) Stream length gradient equation (c) Water-balance equation (d) Darcy equation ===== PAGE 4 ===== 4

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**137. [UNKNOWN] unknown**

9. Crevasses form along the: (a) Margins of glaciers in places where the path is curved (b) Bed rock, by the marking of



boulders that are dragged at the base of ice sheet (c) Grinding of rocks into a powder of fine fragment of silt and clay size (d) Marking on rock in the form of scratches by the sharp corner of rock fragments

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**138. [UNKNOWN] unknown**

10. The threshold flow velocity at which a particle settles to a channel bed is known as: (a) Fall velocity (b) Increasing velocity (c) Decreasing velocity (d) Runoff velocity

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**139. [UNKNOWN] unknown**

11. Pedalfer is a type of soil in the wet climate, it is formed by: (a) An effective downward leaching due to the high rainfall and acids produced by the decay of abundant humus (b) An upward movement of soil water beneath the land surface characterized by little leaching, scant humus (c) The cementation of clay particles, silica and iron compound that have accumulated at ■ horizon (d) An intense and deep weathering in tropical regions where silica disintegrates into clay and mica ===== PAGE 5 ===== 5

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**140. [UNKNOWN] unknown**

12. Abrasion is a process that can mechanically weather rock, and: (a) Solvent action of water which attack in the mineral structure for disintegration (b) Disintegration of rock masses by the process of frost wedging (c) Exposing fresh country rock to the surface due to the removal of their overlying weathering material (d) Grinding away of rocks by friction and impact during transportation

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**141. [UNKNOWN] unknown**

13. If the beds are vertical, the synclinal axis is in the: (a) Opposite direction from that in which the bedding dips (b) Opposite direction from that in which the cleavage dips (c) Same direction from that in which the bedding dips (d) Same direction from that in which the cleavage dips

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**142. [UNKNOWN] unknown**

14. In a plunging fold, if the strike of cleavage is diagonal to the strike of the bedding, the plunge of fold can be determined by: (a) Obtuse angle between strike of bedding on vertical plane and strike of cleavage on horizontal plane (b) Acute angle between line drawn on the ground perpendicular to the strike of bedding and strike of cleavage (c) Obtuse angle between strike of bedding and cleavage strike (d) Acute angle between the line drawn parallel to the strike and dip direction of cleavage ===== PAGE 6 ===== 6

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**143. [UNKNOWN] unknown**

15. With regard to strain-slip cleavage, in metamorphic terranes the schistosity may be crinkled into: (a) Small wavelength folds, and the mica flakes are rotated into non discrete zones (b) Big wavelength folds, and the mica flakes are rotated into non discrete zones (c) Big wavelength folds, and the mica flakes are rotated into discrete zones (d) Small folds with a wavelength of a fraction of an inch and the mica flakes are rotated into discrete zones

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**144. [UNKNOWN] unknown**

16. The slip refers to the displacement along the fault line, while shift is used to refer to: (a) An angle between the fault plane and vertical plane that strikes parallel to the fault (b) The differential movement takes place by systematic small displacement along a large number of closely spaced fractures (c) The total displacement measured on fault surface between two adjacent points situated on opposite walls of fault (d) The displacement on opposite site of fault and outside the dislocated zone ===== PAGE 7 ===== 7

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**145. [UNKNOWN] unknown**

17. Consider the following regarding apparent dip and true dip:  $\theta$  is the apparent dip on a vertical plane  $\phi$  is the angle between strike of bed and direction of apparent dip  $\delta$  is the true dip Which one of the following is the correct equation for an apparent dip of a bed in any desired direction from the true dip? (a)  $\tan \theta = \tan \phi \sin \phi$  (b)  $\tan \theta = \tan \phi \sin \theta$  (c)  $\tan \theta = \tan \phi \sin \delta$  (d)  $\tan \theta = \tan \phi \sin \theta$

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**146. [UNKNOWN] unknown**

18. Consider the following regarding Mohr stress envelope:  $c$  is the cohesive strength of the rock  $\phi$  is the angle of internal friction  $\sigma$  is the normal stress across the plane  $\tau$  is the shear strength on the failure plane The curve of the Mohr envelope is determined by which one of the following equations? (a)  $\tau = c + \sigma \tan \phi$  (b)  $\tau = c + \sigma \tan \phi$  (c)  $\tau = c + \sigma \tan \phi$  (d)  $\tau = c + \sigma \tan \phi$  ===== PAGE 8 ===== 8

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**147. [UNKNOWN] unknown**

19. The space lattice of a mineral is: (a) An assemblage of atoms that is constantly repeated to form a mineral. (b) A glide plane parallel to which there are number of additional planes along which gliding takes place (c) A position of mineral in terms of their resistance towards failure (d) A position of mineral before plastic deformation and recrystallization

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**148. [UNKNOWN] unknown**

20. A tectonite is: (a) A recrystallized rock without any change in shape and orientation of grain (b) A rock showing an imperfection in crystal structure resulting reorientation of rock in terms of volume not in structure (c) A rock that result from the accumulation of many separate components and each of which moved into place independently of its neighbors (d) A deformed rock, the fabric of which is due to the systematic movement of the individual units under a common external force

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**149. [UNKNOWN] unknown**

21. The topography has no control on the outcrop pattern of vertical beds whereas the top and bottom of a vertical bed will appear on the map as: (a) Vertical lines perpendicular to the bed (b) Vertical lines parallel to the dip direction (c) Straight lines parallel to the strike of the bed (d) Straight line perpendicular to the bed ===== PAGE 9 ===== 9

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**150. [UNKNOWN] unknown**

22. In the Rule of  $\theta$ , the outcrop pattern of beds that dip: (a) Downstream at an angle greater than the stream gradient forms a  $\theta$ , the apex of which points downstream (b) Downstream at an angle greater than the stream gradient forms a  $\theta$ , the apex of which points upstream (c) Downstream at an angle smaller than the stream gradient forms a  $\theta$ , the apex of which points downstream (d) Downstream at an angle smaller than the stream gradient forms a  $\theta$ , the apex of which points upstream

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**151. [UNKNOWN] unknown**

23. In  $\theta$ , the projections of the poles which are perpendicular to a horizontal plane will coincide: (a) With the poles of diametrically opposite (b) At the  $\theta$ - $\theta$  quadrant of the net (c) With the center of the projection (d) At the  $\theta$ - $\theta$  quadrant of the net

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**152. [UNKNOWN] unknown**

24. The Chromite deposits in the north limb of the Sukinda syncline in Orissa are blanketed by: (a) Breccia and

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**153. [UNKNOWN] unknown**

25. ■■■■ ■■ deposits are: (a) Greyish-black variety of corundum, containing much admixed magnetite and hematite (b) Red gemstone of corundum found in metamorphosed bauxite deposit (c) Placer variety of corundum having rounded pebbles in alluvium (d) Thermally altered alumina rich shales in zenothic of basic igneous rock

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**154. [UNKNOWN] unknown**

26. A tetragonal unit is a right square prism with ■ and ■ edge lengths equal and ■ different with two space lattices: (a) Primitive and body centered (b) Primitive and face centered (c) Primitive only (d) Body centered and face centered

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**155. [UNKNOWN] unknown**

27. In feldspar group of minerals, while cooling, unmixing of potash feldspars and soda feldspars takes place. The larger component occurs as ground mass and the minor one oriented parallel lamellae within the groundmass. This process of unmixing of two minerals is: (a) Exsolution (b) Isomorphism (c) Polymorphism (d) Magmatic segregation ===== PAGE 11 ===== 11

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**156. [UNKNOWN] unknown**

28. Polymorphism is the phenomenon when a mineral exhibits: (a) Distinctly different physical or optical properties in spite of possessing the same chemical composition (b) Identical physical or optical properties in spite of possessing the same chemical composition (c) Identical physical or chemical properties in spite of possessing the different crystal lattice (d) Distinctly different physical or chemical properties in spite of possessing the same crystal lattice

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**157. [UNKNOWN] unknown**

29. The characteristic feature of a ■■ ■■■■■■■■ ■■■■ group is: (a) 3.5 oxygens are shared between adjacent tetrahedra (b) 2 oxygen atoms link together to form a ring of composition (c) 2 tetrahedra sharing common oxygen (d) 2.5 oxygens are shared by adjacent tetrahedra

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**158. [UNKNOWN] unknown**

30. The characteristic feature of an ■■■■ is: (a) It is contaminated form of silica with varying amount of ■■ and ■■ in its structures (b) It contains a variable quantity of water molecules in its structures (c) It contains large numbers of micro-cavity in its structure (d) It contains traces of colloidal carbon and iron in its structure ===== PAGE 12 ===== 12

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**159. [UNKNOWN] unknown**

31. The fibrous variety of anthophyllite is called: (a) Glaucophane (b) Jadeite (c) Enstatite (d) Amosite

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**160. [UNKNOWN] unknown**

32. Piezoelectricity is the simultaneous development of: (a) Positive and negative charges of electricity when its temperature is suitably changed (b) Positive and negative charges of electricity on different parts of same crystal when its pressure is suitably changed (c) Magnetic charge when its pressure and temperature is suitably changed (d) Electrical charges on different parts of same crystal when its pressure is suitably changed

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**161. [UNKNOWN] unknown**

33. ■■■■ ■■■■ ■■■■ ■■■■ minerals are those which have: (a) An uniform white streak though the colour is variable (b) A different colour based on chemical composition (c) A characteristic and constant colour (d) The same physical appearance in spite of having different chemical composition ===== PAGE 13 ===== 13

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**162. [UNKNOWN] unknown**

34. ■■ ■■■■ ■■■■ consisting mainly of an: (a) Alkali feldspar host with an exsolved ■■-rich plagioclase feldspar phase which segregated from the host during cooling (b) ■■-rich feldspar as host with an exsolved alkali feldspar segregated from host (c) Equal amount of alkali feldspar and a more ■■-rich plagioclase (d) Intrusion of ■■-rich plagioclase into alkali feldspar with a phase of segregation

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**163. [UNKNOWN] unknown**

35. Duhem's Theorem states that, a system is said to be closed if: (a) No masses are added to or subtracted from it in the course of the transformation under consideration (b) Masses are added to it in the course of the transformation under consideration (c) Masses are subtracted from it in the course of the transformation under consideration (d) Masses are partially added to or subtracted from it in the course of the transformation under consideration

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**164. [UNKNOWN] unknown**

36. If crystallization is strictly along Bowen's reaction series, such crystallization is called: (a) Cotectic crystallization (b) Eutectic crystallization (c) Peritectic crystallization (d) Fractional crystallization ===== PAGE 14 ===== 14

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**165. [UNKNOWN] unknown**

37. The ■■■■ classification of plutonic rocks is applicable to visually crystalline (Phaneritic) rocks which are presumed to have formed at considerable depth. This classification is strictly: (a) Based on silica percentage (b) A quantitative mineralogical (c) A depth wise (d) Based on texture

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**166. [UNKNOWN] unknown**

38. ■■■■ ■■■■■■■■ ■ are: (a) Calciferous mineral rich in silica than feldspar (b) Aluminosilicates which occur as constituents of igneous rocks and which are poorer in silica than corresponding alkali feldspars (c) Monomineralic rocks which only composed of sodic feldspar (d) Aluminosilicates saturated with silica and poorer in alkali feldspars

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**167. [UNKNOWN] unknown**

39. ■■ ■■■■ ■■ structure is a special type of orbicular structure where ovoid phenocrysts of ■-feldspar are mantled by shells of granular oligoclase. This structure is typically developed in: (a) Merozone granites (b) Catazone granites (c) Epizonal granites (d) ■-type granites ===== PAGE 15 ===== 15

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**168. [UNKNOWN] unknown**

40. ■ ■■■■■■ ■ texture is a type of quench texture and resembles a bunch of 'grass'. It is usually associated with: (a) Low magnesian lava (b) Granitic lava (c) High magnesian lava (d) Andesitic lava

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**169. [UNKNOWN] unknown**

41. Incongruent melting occurs when: (a) A melt break down into two or more than two chemically distinct phase by assimilation (b) A solid split at very high temperature into two solids having same bulk chemical composition (c) A solid splits at definite temperature into another solid and liquid (d) A rock is subjected to high confining pressure without any

increase in temperature

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**170. [UNKNOWN] unknown**

42. Influence of dissolved water is one of the process of magmatic differentiation, it involves: (a) Alkalis and certain metals co-ordinate with the water and concentrate in the regions of lowest pressure and temperature (b) Alkalis and metals are hydrophobic so they move from water saturated area to water deficient area (c) Different chemical potential of water throughout the magma chamber (d) Water provides a way to gravitational rising of heavy crystal ===== PAGE 16 ===== 16

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**171. [UNKNOWN] unknown**

43. During the complex process of reaction between magma and invaded rock, magma becomes modified by incorporation of material originally present in wall rock. This process is termed as magmatic: (a) Differentiation (b) Mixing (c) Assimilation (d) Crystallization

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**172. [UNKNOWN] unknown**

44. Kimberlites are alkalic rocks characteristically ultramaphic in composition. They differ from lamprophyres and lamproites in: (a) Kimberlites have high silica (b) Contents of alkalis i.e.  $\text{Na}_2\text{O}$  and  $\text{K}_2\text{O}$  in Kimberlites are in a lower level (c) Contents of  $\text{MgO}$  are low in Kimberlites (d) Contents of alkalis are high in Kimberlites

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**173. [UNKNOWN] unknown**

45. Crystallization may be localized at a cooling margin, where the temperature is lower than the central parts of magma, thus: (a) A solid phase is formed (b) A liquid phase is formed (c) A solid and a liquid phases are formed (d) A Liquid and gaseous phases is formed ===== PAGE 17 ===== 17

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**174. [UNKNOWN] unknown**

46. A certain mineralogical change in rock that is also of local extent but is of different character from either contact or cataclastic metamorphism is called: (a) Cataclastic metamorphism (b) Hydrothermal metamorphism (c) Dynamo-thermal metamorphism (d) Regional Burial metamorphism

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**175. [UNKNOWN] unknown**

47. Nucleation in metamorphic rocks is: (a) Faster when entropy change is smallest (b) Faster when entropy change is highest (c) Initiated by entropy but not depends on entropy in advance stage (d) Controlling factor for second order transitions

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**176. [UNKNOWN] unknown**

48. Consider the following regarding phase rule:  $F$  is the number of independent degrees of freedom for the system  $C$  is the number of chemical components  $P$  is the number of phases The phase rule is expressed by which one of the following equations? (a)  $F + P = C + 2$  (b)  $F + C = P + 2$  (c)  $F + C = P - 2$  (d)  $F + P = C - 2$  ===== PAGE 18 ===== 18

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**177. [UNKNOWN] unknown**

49. A foliation marked by the parallel orientation of tabular minerals in a metamorphic rock with a sufficiently coarse grain size is called a: (a) Cleavage (b) Schistosity (c) Gneissosity (d) Solution cleavage

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**178. [UNKNOWN] unknown**

50. Pseudotachylytes are: (a) Thin seams of glassy igneous intrusion by extreme from the magma with mafic composition (b) Thin seams of glassy rock produced of the results of melting due to frictional heating and are restricted to dry rocks (c) A rock mass with random fabric of interlocking with a tough texture and strength (d) Fine grained rock produced in the zone of intense ductile deformation whereas pre existing grains have been recrystallized as finer grains

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**179. [UNKNOWN] unknown**

51. When an igneous rock is metamorphosed and the original texture continues to occur in metamorphosed ones, the resulting texture is said to be: (a) Granoblastic texture (b) Blasto-porphyritic texture (c) Porphyroblastic texture (d) Xenoblastic texture ===== PAGE 19 ===== 19

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**180. [UNKNOWN] unknown**

52. Metamorphic facies can be defined as: (a) A range of chemical reaction under the process of metamorphism where bulk chemical composition of altered rock remain same as parent rock (b) A range of pressure-temperature conditions over which a particular common mineral assemblage or range of mineral assemblages are stable (c) A physically separable constituents of the system in the form of solid, liquid or gas (d) A changing condition of pressure and temperature which lead to reaction and production of new assemblage

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**181. [UNKNOWN] unknown**

53. A specific change in mineral assemblage reflecting a metamorphic reaction and satisfying the condition of reaction equilibrium is called: (a) Isograd (b) Isopatch (c) Isocontour (d) Line of phase change

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**182. [UNKNOWN] unknown**

54. Skarns are: (a) Metasedimentary rock that are rich in ■■■-or ■■■-■■■ silicate (b) Recrystallized rock where all the igneous minerals recrystallized into high grade metamorphic minerals (c) Metamorphic rocks formed by hydrothermal alteration of limestone into marble (d) A variety of calc-silicate rock formed by metasomatic interaction between marble and silicate rock ===== PAGE 20 ===== 20

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**183. [UNKNOWN] unknown**

55. In a ■■■ ■■■■ type paired metamorphic belt: (a) The high and low pressure belt of the same age occurred next to one another, separated by major faults (b) Two high pressure belts of the different age occurred next to one another (c) The high and low pressure belt of different age occurred next to one another separated by faults (d) Two low pressure belts of the different age occurred next to one another

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**184. [UNKNOWN] unknown**

56. The steps involved in a metamorphic reaction will be the slowest in a natural geological setting and will therefore control the rate of the overall reaction is known as: (a) Rate limiting step (b) Surface step (c) Heat energy step (d) Natural reaction step

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**185. [UNKNOWN] unknown**

57. The difference between carbonate and siliciclastic sediment is: (a) The carbonate sequence occur in shallow tropical environments while siliciclastic occurs worldwide and at all depths (b) The carbonate sequence occur at deep marine environments while siliciclastic occurs at all depths (c) Both carbonate and siliciclastic occur at same depositional environment with only difference of grain size and texture (d) The carbonate deposits results from shallow water bodies from interaction of current and waves while siliciclastic sediments are result of only deep, calm water deposit =====

58. Grain size interpretation in siliciclastic rock is much simpler than carbonate and evaporites because: (a) Diagenesis in carbonates and evaporites obscures original grain relationship (b) Siliclastic rocks are cemented on sea floor and easy for stratigraphic co-relation (c) For carbonate and evaporites climate is no constraints, sediment occur worldwide and at all depth (d) In carbonate sediment periodic exposure of sediment during deposition leaves deposits relatively unaffected

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59.     are: (a) Microscopic erosional surface on the underside of bedding plane formed by natural casts of erosional feature (b) Vortex erosion typically at the base of turbidity current (c) Organically bored surfaces which encrusted with fossils in growth position (d) Resistant surface altered by erosional impact of large objects entrained in the flow

Official source: [backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf](https://backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf)

60. In sedimentary rocks, the beds that contain internal layers that are essentially parallel to the bounding bedding surfaces are said to be: (a) Cross stratified (b) Planar stratified (c) Composite bedsets (d) Laminated bedding =====  
PAGE 22 ===== 22

Official source: [backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf](https://backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf)

61. Ichnofacies dominated by deep vertical burrows in a tidal zone named the: (a) Scolithos (Skolithos) (b) Cruziana (c) Zoophycus (d) Nereites

Official source: [backend/data/pyg/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf](https://backend/data/pyg/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf)

62. The clastic ratio of a lithofacies is calculated by: (a)  (b)   
 (c)  (d)

Official source: [backend/data/pyg/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf](https://backend/data/pyg/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf)

63. Textural maturity of sandstone is determined by: (a) The relative abundance of stable and unstable framework grains (b) The ratio of matrix and well defined grain under hydrostatic condition (c) The relative abundance of matrix and the degree of rounding and sorting of framework grains (d) The proportion of coarse grain sediment to fine grain sediment when matrix is absent ===== PAGE 23 ===== 23

Official source: [backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf](https://backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf)

64. Vertical profile model of a cyclic sequence in evaporite bearing sediments, reflecting a sensitive response of evaporite environments to: (a) Climate change, brine level or water chemistry (b) Change in lateral dispersal of sediment as sea level changes (c) Vertical profile in evaporites cyclic sequence does not reflect (d) Change in sediment supply as a response of faunal and floral intervention

Official source: [backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf](https://backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf)

**193. [UNKNOWN] unknown**

65. The non-clastic sediment composed of microcrystalline quartz, found in both bed and nodules of limestones and sandstones is known as: (a) Cherts (b) Silty sandstone (c) Breccia (d) Evaporites

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**194. [UNKNOWN] unknown**

66. Which one of the following zone of carbonate diagenetic depth occurs below the calcite compensation depth ■■■■■? (a) Zone of precipitation (b) Zone of partial dissolution (c) Zone of no carbonate (d) Zone of active dissolution

===== PAGE 24 ===== 24

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**195. [UNKNOWN] unknown**

67. Which one of the following may be caused by carbonic acid-rich waters produced by the devolatilization of coal beds? (a) Kaolinization (b) Montmorillonite (c) Rudaceous rock formation (d) Salt rock formation

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**196. [UNKNOWN] unknown**

68. Reef and carbonate slope build-ups have steeply dipping sediment-water interfaces, much steeper than the angle of repose of loose talus, is due to: (a) Deposits at equatorial regions only and grows towards overhead sun (b) Characteristic feature that grows vertical or overhanging (c) Organic binding and early cementation (d) Vertical growth from sea beds

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**197. [UNKNOWN] unknown**

69. The most favourable conditions for fossilization are present in: (a) Terrestrial (b) Fluvial (c) Sea (d) Lacustrine

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**198. [UNKNOWN] unknown**

70. Fossils range from ancient time upto the present day without any change in their primitive characters are known as: (a) Living fossils (b) Derived fossils (c) Facies fossils (d) Relict fossils ■fossils ===== PAGE 25 ===== 25

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**199. [UNKNOWN] unknown**

71. A specimen of a species from the same locality as its holotypes or syntypes is a: (a) Plesiotype (b) Metatype (c) Paratype (d) Topotype

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**200. [UNKNOWN] unknown**

72. Two populations differing in one or more morphologic characteristics to a statistically significant degree are called: (a) Morphonomical species (b) Morphospecies (c) Founder species (d) Allopatric species

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**201. [UNKNOWN] unknown**

73. Which one of the following is the median part of the cephalon that inflated in trilobites? (a) Glabella (b) Cheeks (c) Axial furrows (d) Pleuron ===== PAGE 26 ===== 26

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**202. [UNKNOWN] unknown**

74. Which one of the following is the oldest known ammonoides during upper Silurian–Devonian period? (a) Trilobites (b) Gymmites (c) Agoniatites (d) Prolecanites

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf



**203. [UNKNOWN] unknown**

75. In Gastropods, the cavity opens at the base of the shell is called: (a) Umbilicus (b) Columella (c) Whorl (d) Apex

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**204. [UNKNOWN] unknown**

76. A straight conical shell of an ammonoid cephalopod is known as: (a) Camerae (b) Cyrotocone (c) Bactriticone (d) Roastrum

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**205. [UNKNOWN] unknown**

77. The earliest fossil remains of horse in equine evolution is: (a) Epihippus (b) Mesohippus (c) Eohippus (d) Miohippus  
===== PAGE 27 ===== 27

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**206. [UNKNOWN] unknown**

78. The modern horses belong to which one of the following genus? (a) Epihippus (b) Equus (c) Miohippus (d) Orohippus

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**207. [UNKNOWN] unknown**

79. The first formed chamber in multilocular form of foraminifera is known as: (a) Protegulum (b) Proloculum (c) Protoconch (d) Prodissononch

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**208. [UNKNOWN] unknown**

80. In which one of the following rhizome (root) on which Glossopteris and Gangamopteris grew? (a) Ptilophyllum (b) Vertebraria (c) Calamites (d) Lepidodendron

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**209. [UNKNOWN] unknown**

81. In conformable contacts, if the change from one lithology to another is less marked than abrupt contacts, reflecting gradual change in depositional conditions with time is said to be: (a) Gradational (b) Lateral Adjustment (c) Lithostratigraphy (d) Angular Unconformity  
===== PAGE 28 ===== 28

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**210. [UNKNOWN] unknown**

82. When geological formations are equivalent in age but situated wide apart in different districts, countries or continents, they are said to be: (a) Homotaxis (b) Homotaxial (c) Synchronous beds (d) Contemporaneous

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**211. [UNKNOWN] unknown**

83. The Sargur Schist Complex rocks have undergone a high degree of metamorphism ranging from amphibolites to lower granulite facies, and these rocks have often been extensively migmatized and mobilized during the subsequent phase of emplacement of: (a) Peninsular Gneissic Complex (b) Closepet Granite (c) Bundelkhand Granite (d) Newer Green Stone Belt

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**212. [UNKNOWN] unknown**

84. The characteristic feature of the '■■■■ ■■■■■■ ■■ ■■ ■■■■ ■' of Eastern Ghats Province is: (a) Hypersthene bearing granite composed of blue quartz, felspar and hypersthene (b) Garnetiferous quartzite (c) Potash felspar with graphite and manganese (d) Dry metamorphic rocks comprising minerals devoid of water  
===== PAGE 29 ===== 29

**213. [UNKNOWN] unknown**

85. The gneisses known as '■■■■■■■■■■' belonging to '■■■■■■■■■■ ■■ ■■ ■■■ ■' are rich in: (a) Iron ore (b) Manganese ore (c) Copper (d) Lead-Zinc

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**214. [UNKNOWN] unknown**

86. The Cuddapah system is made up of alternate layers of: (a) Shales and sandstones (b) Quartzites and shales (c) Shales and breccia (d) Quartzite and breccia

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**215. [UNKNOWN] unknown**

87. The Vindhyan system lies unconformably on the: (a) Kurnool rocks (b) Kuling rocks (c) Cuddapah rocks (d) Kanwar rocks

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**216. [UNKNOWN] unknown**

88. The ■■■■■■■■ ■■■■ rocks of fresh water origin which occur in Madhya Pradesh are called the: (a) Lameta group (b) Bagh bed (c) Nimar sandstone (d) Umia sandstone ===== PAGE 30 ===== 30

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**217. [UNKNOWN] unknown**

89. A series of mainly acidic volcanic rocks belonging to the '■■■■■ ■■■■■■ ■■■■■■ ■■■■■■' are known as: (a) Rialo series (b) Malani Volcanics (c) Charnockite series (d) Khondalite series

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**218. [UNKNOWN] unknown**

90. Trigonina beds with the characteristic fossils of Trigonina of Mesozoic succession of Kutch Region belongs to: (a) Chari Formation (b) Patcham Formation (c) Umia Formation (d) Katrol Formation

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**219. [UNKNOWN] unknown**

91. Which one of the following basins of the Damodar Valley region shows the maximum development of coaliferous beds that contain several superior quality coal measures? (a) Barker basin (b) Talchir basin (c) Raniganj basin (d) Panchet basin ===== PAGE 31 ===== 31

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**220. [UNKNOWN] unknown**

92. The origin and structure of Indo-Gangetic Plain are closely related with the rise of the Himalaya. This plane is: (a) A basin caused by the central rifting of continental crust during Himalayan orogeny (b) A dome of Precambrian rock with representation of the sediment in the same basin (c) A plateau with the Archean Basement unconformably overlain by the sediment of Tertiary sediment (d) A deep crustal trough filled with Quaternary sediments

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**221. [UNKNOWN] unknown**

93. The '■■■■■■■ ■■■■■■' rocks were deposited in: (a) A geosynclinal basin bordering the shelf sea of the Zaskar-Spiti region (b) A shallow freshwater stream flowing in a wide sedimentary basin (c) A deep sea with phases of sea transgression and regression (d) A deep lake deposit with the intervening sea fringes

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**222. [UNKNOWN] unknown**

94. A process of mineral deposits associated with volcanism, thermal springs and fumaroles wherefrom volatilized matter is redeposited at lower temperature and pressure is: (a) Evapotranspiration (b) Oxidation (c) Sedimentation (d) Sublimation ===== PAGE 32 ===== 32

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**223. [UNKNOWN] unknown**

95. Tenor is: (a) The concentration of gang mineral associated with the ore mineral (b) The metal content of an ore, expressed in percent (%), and is expressed in ounces per tonne in case of precious metals or gems (c) The total quantity of matter that is used to extract a particular kind of ore (d) The ratio of total quantity of host rock to total quantity of ore extracted

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**224. [UNKNOWN] unknown**

96. Most of the primary gold deposits occur in the vicinity of: (a) Basic igneous intrusions (b) Acidic igneous intrusions (c) Intermediate igneous intrusions (d) Pegmatite igneous intrusions

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**225. [UNKNOWN] unknown**

97. The rock that holds the oil and gas in place so that they do not escape until released by drilling is known as: (a) Reservoir rock (b) Cap rock (c) Basement rock (d) Stratigraphic trap ===== PAGE 33 ===== 33

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**226. [UNKNOWN] unknown**

98. Auriferous veins occur in the Hutti-Gold-field within the schistose rocks of: (a) Permian age (b) Cambrian age (c) Dharwarian age (d) Archean age

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**227. [UNKNOWN] unknown**

99. '■■■■■■■ ■■■■ ■■■■■■ - ■■■■■■■■■■' deposit of Rajasthan show the mineralogical changes in wall rock brought about by the depletion of some element from the wall rock. These deposits are: (a) Epigenetic (b) Syngenetic (c) Postgenetic (d) Residual

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**228. [UNKNOWN] unknown**

100. The '■■■■■ ■■■■ ■' represent: (a) A gaseous phase where substances are carried and demonstrate the ability of gaseous emanations to collect, transport and deposit the metal (b) A water phase contain many mineral substance with higher temperature water dissolve, transport and deposit mineral (c) A magmatic source of mineral deposits which transport and deposit many non metallic mineral as hydrothermal injections (d) A mineral deposits of copper and lead formed by supergene enrichment after a volcanic eruption ===== PAGE 34 ===== 34

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**229. [UNKNOWN] unknown**

101. The cap-rock of an ore deposit in the form of cellular mass of limonite and gangue formed due to oxidation of an ore and points to the nature of underlying hidden deposits is called: (a) Carnotite (b) Gossan (c) Host rock (d) Parent rock

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**230. [UNKNOWN] unknown**

102. The process of metasomatic replacement is by: (a) Alteration (b) Hydrolysis (c) Hydrostatic alteration (d) Diffusion

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

103. Oil formed from marine organisms buried in sediments is explained by: (a) Animal-origin theory (b) Vegetable-origin theory (c) Mendeleef's carbide theory (d) Cosmic theory ===== PAGE 35 ===== 35

104. Gold province of Karnataka-Andhra Pradesh-Tamil Nadu is occupied by Dharwars, the rock type in the area are represented by: (a) Granite that intruded schist (b) Metamorphosed mafic volcanic rocks altered to schistose, granular and fibrous amphibolites (c) Precambrian tremalite, actinolite schist and granite gneiss (d) Metamorphosed flysch with exogenic quartzite and limestone

105. In an area of 100 ■■■, the water table dropped by 4.5 ■, the porosity is 30 % and specific retention is 10 %. The change in groundwater storage will be: (a) 3 ■10■■■■ (b) 6 ■10■■■■ (c) 9 ■10■■■■ (d) 12 ■10■■■■

106. Which one of the following types of rocks are relatively high porous because there is a great variation in degree of packing on the one hand and in size, shape and arrangement of the grains on the other hand? (a) Igneous (b) Metamorphic (c) Sedimentary (d) Crystalline rocks ===== PAGE 36 ===== 36

107. In designing a critical underpass drain to accept a 10 % risk that flooding will occur in the next 5 ■■■ ■■. The return period for a highway engineer use will be nearly: (a) 58 ■■■ ■■ (b) 52 ■■■ ■■ (c) 48 ■■■ ■■ (d) 42 ■■■ ■■

108. Consider the following regarding Darcy's law:  $L$  is the length of the column  $K$  is the hydraulic conductivity  $\Delta h$  is the difference in hydraulic head Which one of the following is the correct rate of flow or flux velocity of groundwater  $q$  through a column of saturated sand in accordance with the Darcy's law? (a)  $q = \frac{K \Delta h}{L}$  (b)  $q = \frac{L \Delta h}{K}$  (c)  $q = \frac{K}{L \Delta h}$  (d)  $q = \frac{L}{K \Delta h}$

109. The Sodium adsorption ratio  $\frac{C_{Na}}{C_{Ca} + C_{Mg}}$  is: (a)  $\frac{C_{Na}}{C_{Ca} + C_{Mg}}$  / (b)  $\frac{C_{Na}}{C_{Ca} + C_{Mg}}$  / (c)  $\frac{C_{Na}}{C_{Ca} + C_{Mg}}$  / (d)  $\frac{C_{Na}}{C_{Ca} + C_{Mg}}$  / ===== PAGE 37 ===== 37

110. Which of the following isotopes will be used for an age determination of groundwater? (a) Hydrogen - 3 (Tritium) (b) Uranium (c) Hydrogen - 3 (Tritium) and Uranium (d) Hydrogen - 3 (Tritium) and Carbon - 14

111. Lateral reduction in hydraulic conductivity of aquifer below the recharge area causes a higher hydraulic gradient underneath the recharge basin, adequate to cause: (a) An entire volume of recharged ground water to flow towards the domain boundary (b) The hydraulic conductivity of the function to reduce at the centre of aquifer (c) The value of groundwater head to remain constant for a section in plain surface (d) The ground water head value to a lower level at margin

Official source: <backend/data/pyg/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf>

**240. [UNKNOWN] unknown**

112. Consider the following regarding surface depressions:  $S$  is the maximum storage capacity of the depressions  $R$  is the rainfall excess  $K$  is constant Which one of the following is the correct volume  $V$  of water stored by surface depressions at any given time? (a)  $V = \frac{1}{2} S R$  (b)  $V = \frac{1}{2} S K$  (c)  $V = \frac{1}{2} R K$  (d)  $V = \frac{1}{2} S K$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**241. [UNKNOWN] unknown**

113. Consider the following regarding an unconfined aquifer:  $S_y$  is the average specific yield of the aquifer  $\Delta V$  is the volume of the aquifer lying between the original water table and the water table at some later specific time Which one of the following is the correct change in storage volume  $\Delta V$  in an unconfined aquifer? (a)  $\Delta V = S_y \Delta V$  (b)  $\Delta V = S_y \Delta V$  (c)  $\Delta V = S_y \Delta V$  (d)  $\Delta V = S_y \Delta V$

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**242. [UNKNOWN] unknown**

114. Consider the following regarding storage coefficient of the aquifer:  $\gamma_w$  is the specific weight of water  $n$  is the porosity of soil (decimal)  $H$  is the thickness of the confined aquifer  $E_w$  is the bulk modulus of elasticity of water  $E_s$  is the modulus of compressibility (elasticity) of the soil grains of the aquifer Which one of the following is the correct storage coefficient of the aquifer  $S$ ? (a)  $S = \frac{\gamma_w H}{E_s}$  (b)  $S = \frac{\gamma_w H}{E_w}$  (c)  $S = \frac{\gamma_w H}{E_s}$  (d)  $S = \frac{\gamma_w H}{E_w}$

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Directions: -- Each of the next Six (06) items consists of two statements, one labeled as the 'Statement (I)' and the other as 'Statement (II)'. You are to examine these two statements carefully and select the answers to these items using the codes given below: Codes: (a) Both Statement (I) and Statement (II) are individually true and Statement (II) is the correct explanation of Statement (I) (b) Both Statement (I) and Statement (II) are individually true but Statement (II) is NOT the correct explanation of Statement (I) (c) Statement (I) is true but Statement (II) is false (d) Statement (I) is false but Statement (II) is true

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**243. [UNKNOWN] unknown**

115. Statement (I): Material is pulled out of the sun by an external force such as gravitational pull resulting from the dynamic encounter or near collision of the sun with another star. Statement (II): The planets become isolated masses of matter as the material of the solar-system condensed into the sun.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**244. [UNKNOWN] unknown**

116. Statement (I): A strike-slip fault is one in which the net slip is parallel to the strike of the fault. Statement (II): Faults may be classified on the basis of their genesis is called strike slip fault.

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Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**245. [UNKNOWN] unknown**

117. Statement (I): The  $\sigma_1$  diagram is analyzed on the assumptions that the folds are cylindrical. Statement (II): The  $\sigma_1$  diagram involve the projection of the planes rather than perpendicular to the planes.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**246. [UNKNOWN] unknown**

118. Statement (I): Feldspars are the most abundant minerals in the earth crust. Statement (II): Feldspars are essential minerals in only metamorphic rock and stable over a limited range of pressure-temperature conditions.

Official source: backend/data/pyq/collector/universal-official-pdfs/raw/Geol-P-20-Geology-Hydrogeology.pdf

**247. [UNKNOWN] unknown**

119. Statement (I): Igneous rocks are formed during the cooling and hardening of magma within the earth surface. Statement (II): Igneous rocks constitute 65% of the total mass of the rocks composing the crust.

**248. [UNKNOWN] unknown**

120. Statement (I): The texture and mineral composition of sedimentary rocks are of great value in determining the nature of the environment at the time when the sediment was deposited. Statement (II): A conglomerate, arkose and shale indicates high energy, inorganic material, such as swamp and lagoons are examples of sedimentary rocks.